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CO2 - AT1

① A csv file is imported into pandas, but the first row of actual data is wrongly treated as column names. What is the Best correcti -on

Ans: Use header = None while reading the file.

2. A dataset has 25 columns, but only Age, salary and Department are required for analysis. Why should only these columns be sele -ted?

Ans: To reduce unnecessary data processing.

③ A manager wants employees from the IT department with salary above ₹ 60,000. which command is correct?

Ans: df(df['Department'] == 'IT') & (df['salary'] > 60000)]

4. A student dataset is filtered using:

```
df[df['Marks'] >= 50]
```

What does this operation return?

Ans: Students who scored 50 or above.

⑤ A dataset has 40% missing values in one column. What is the m analytical concern before filling missing values?

Ans. Whether missingness may affect the reliability of analysis.

⑥ Mean Imputation is most suitable when:

Ans. The numerical data is approximately symmetric.

⑦ A salary column contains a few very high executive salaries. Some salary values are missing. Which Imputation method is more suitable?

A) Median

8. A city column contains missing values. which method is usually suitable?

A) Replace missing values with the most frequent city.

9. A dataset has 10,000 rows, and only 12 rows contain missing values. what is a reasonable approach?

A) Drop the 12 incomplete rows.

10) A company wants to identify the top-5 highest-paid employees. which sequence is most appropriate?

A) Sort salary in descending order and select first 5 rows.

⑪ A school wants to sort students first by class and then by Marks in descending order within each class. Which command is correct

A) df.sort_values(['class', Marks'], ascending = [True, False])

⑫ A sales dataset contains Region, product, and sales. To find total sales for each region and product combination, which operation is best?

A) df.groupby(['Region', Product']) ['sales'].sum()

(13) A college wants to compare average marks department-wise. Which result best supports this analysis.

A) Mean marks grouped by department.

14. A company wants department-wise average salary and maximum salary. Which command is suitable?

A) df.groupby('Department')['salary'].agg(['mean','max'])

(15) Two students have the same mark. What does ranking help identify?

A) It assigns relative positions while handling ties based on rank meth-od.

(16) A company wants to rank employees by salary within each dept. which command is correct?

A) df['Rank'] = df.groupby('Department')['salary'].rank(ascending = False)

17. A researcher wants to compare average marks of five departments. which visualization is most suitable?

A) Bar chart.

18. A company records monthly profit for 24 months. which chart gives the best analytical understanding of growth or decline?

A) Line chart

19. A dataset contains Advertising Budget and sales Revenue. Which visualization is most suitable to analyze whether higher advertise-ment spending increases sales?

A) Scatter plot.

20. A dataset contains employee salaries, and the analyst suspects some extremely high salaries. Which visualization is most useful?

A) Box plot.